Discussion Exercise 1

University of California, Riverside
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Sense of Scale

1. *Assess whether or not the following statements are reasonable.*

a. *Joe is 180 cm tall.*
b. *I rode my bike to campus at a speed of 50 m/s.*
c. *A skier reaches the bottom of the hill going 25 m/s.*
d. *I can throw a ball a distance of 2 km.*
e. *I can throw a ball at a speed of 50 km/hr.*
f. *Joan's newborn baby has a mass of 33 kg.*
g. *A typical hummingbird has a mass of 3.3 g.*

You can convert the unit system to the one that you are familiar with.

Motion Diagrams and the Particle Model

Exercises 1–5: Draw a motion diagram for each motion described below.

- Use the particle model to represent the object as a particle.
- Six to eight dots are appropriate for most motion diagrams.
- Number the positions in order, as shown in Figure 1.4 in the text.
- Be neat and accurate!

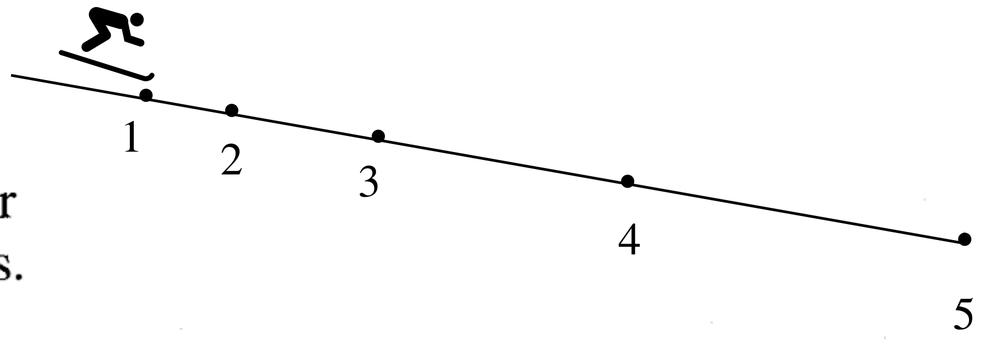
1. A car accelerates forward from a stop sign. It eventually reaches a steady speed of 45 mph.

2. An elevator starts from rest at the 100th floor of the Empire State Building and descends, with no stops, until coming to rest on the ground floor. (Draw this one *vertically* since the motion is vertical.)

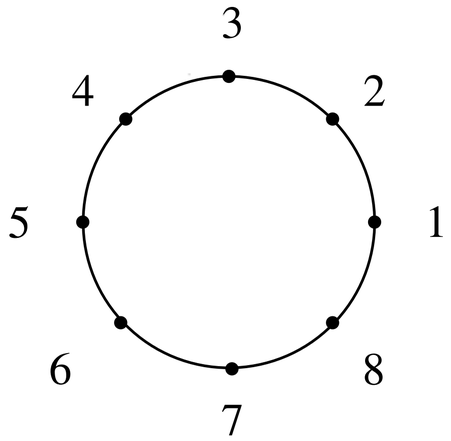
Rotate to make it vertical

Motion Diagrams and the Particle Model

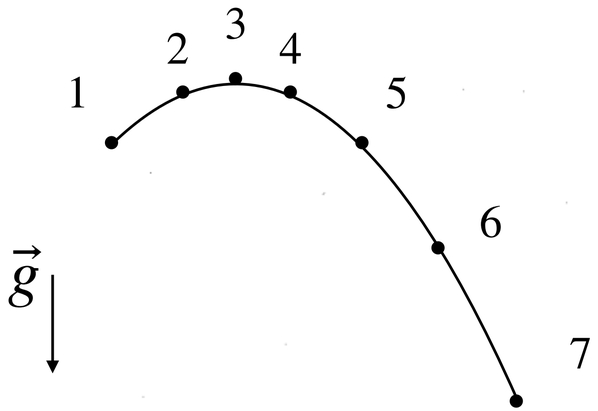
3. A skier starts *from rest* at the top of a 30° snow-covered slope and steadily speeds up as she skies to the bottom. (Orient your diagram as seen from the *side*. Label the 30° angle.)



4. The space shuttle orbits the earth in a circular orbit, completing one revolution in 90 minutes.

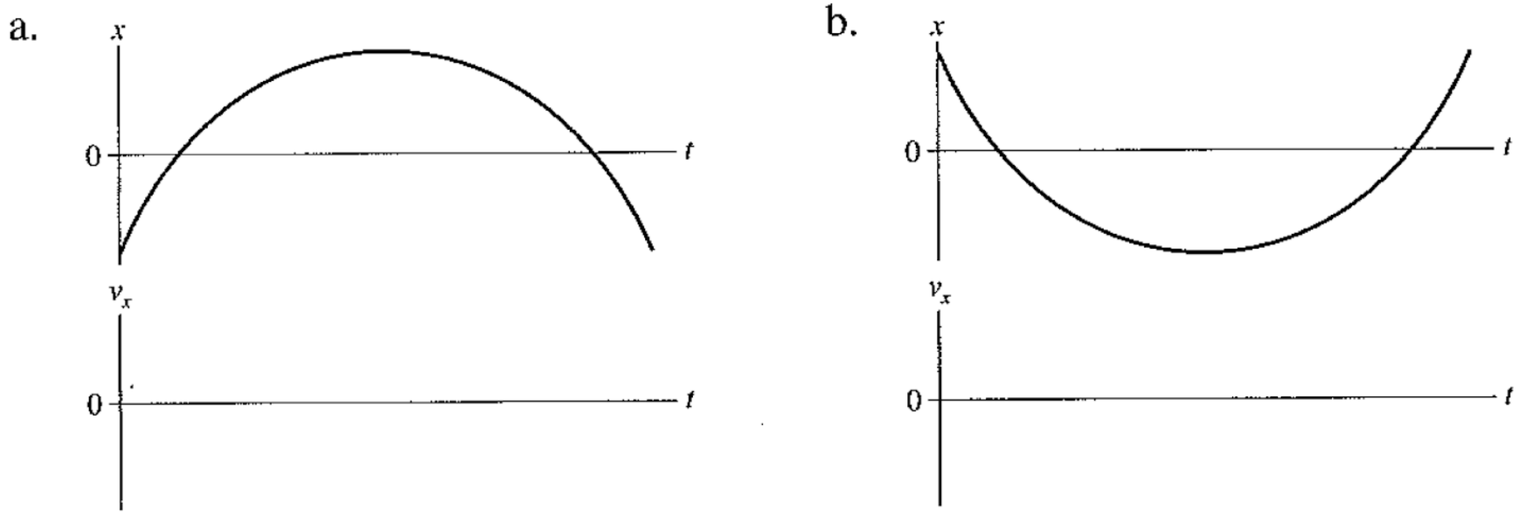


5. Bob throws a ball at an upward 45° angle from a third-story balcony. The ball lands on the ground below.

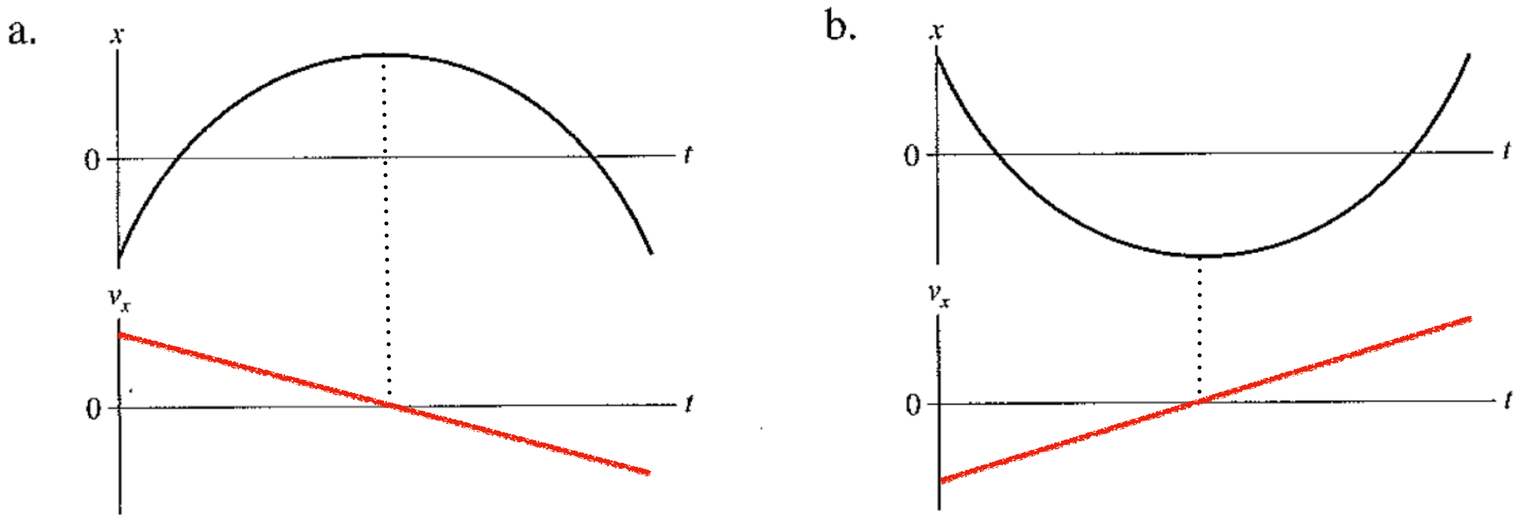


Instantaneous Velocity

Below are two position-vs-time graphs. Redraw each on the whiteboard. For each, draw the corresponding velocity-vs-time graph directly below it. A vertical line through both graphs should connect the velocity v at time t with the position x at the same time t. There are no numbers, but your graphs should correctly indicate the relative speeds.



Instantaneous Velocity



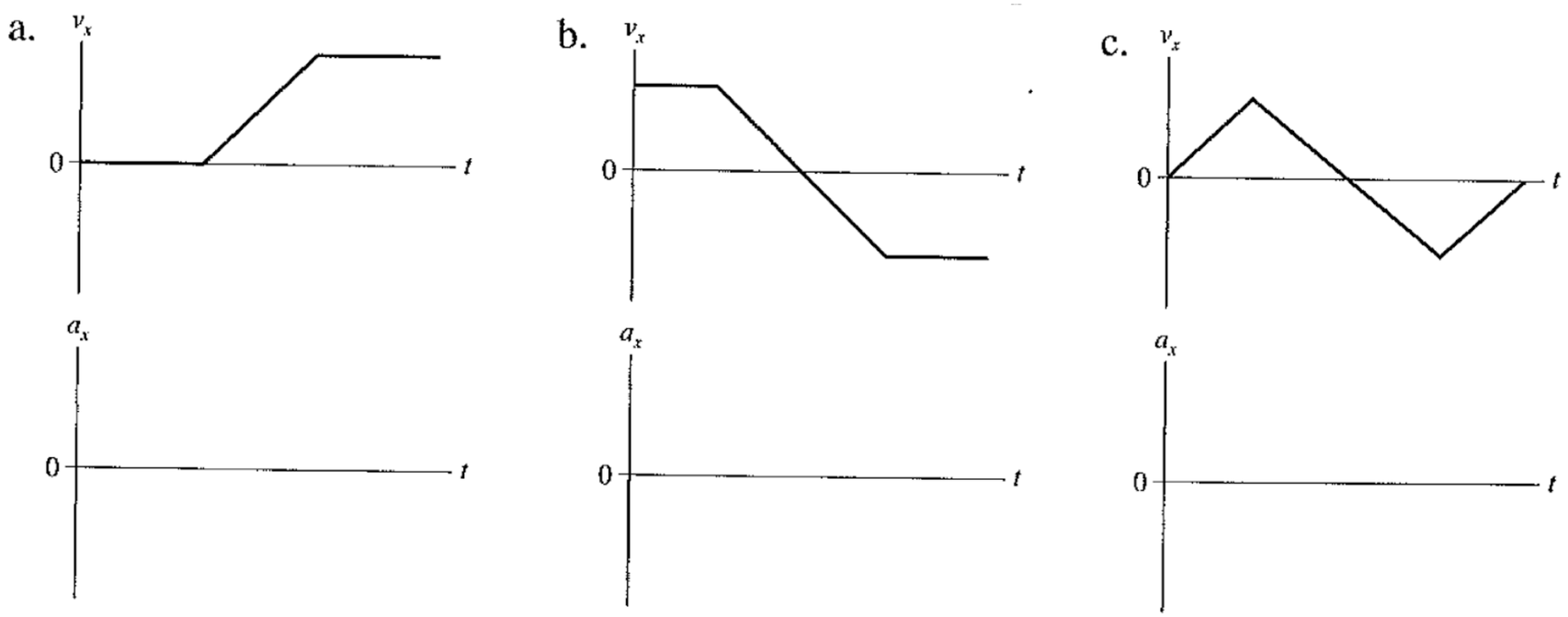
Note: the velocity curve is not necessarily a straight line but should be an increasing (for b)/decreasing function (for a) and the velocity at the middle point should be zero since the slope of x-t graph at that point is zero

Recall the geometry meaning:

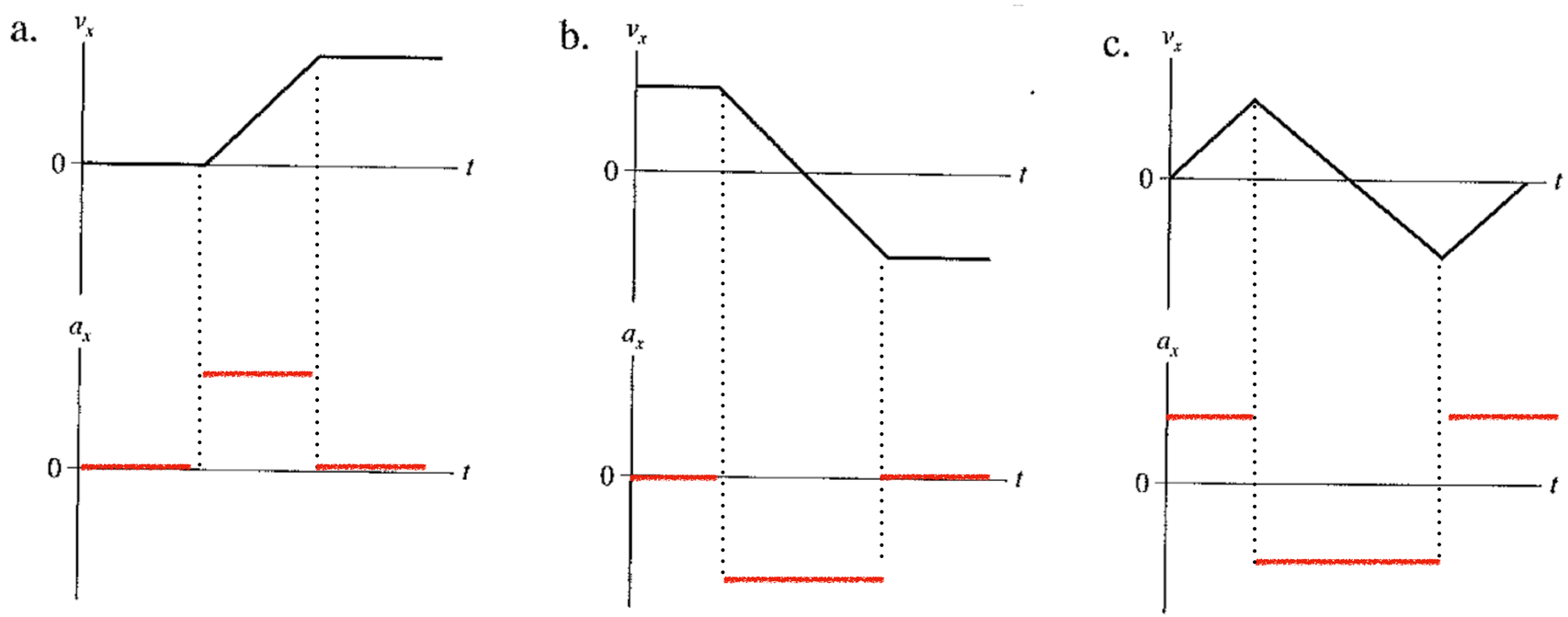
The instantaneous velocity at a point is the slope of the tangential line through that point in the "*displacement v.s. time*"diagram. (Why?)

Motion with constant acceleration

Below are three velocity-vs-time graphs. For each, draw the corresponding acceleration-vs-time graph and draw a motion diagram below the graph.



Motion with constant acceleration

Recall the geometry meaning:

The instantaneous acceleration at a point is the slope of the tangential line through that point in the "*velocity v.s. time*" diagram. (Why?)